South Africa, step 2 (2026-10-20)

chain	seed	iterations.saved	acceptance.rate	final.log.likelihood	best.log.likelihood	errored.proposals
1	20263016	5000	0.276	−34054.9	−34037.4	0
2	20263017	5000	0.276	−34046.5	−34021.0	0
3	20263018	5000	0.276	−34076.8	−34047.6	0
4	20263019	5000	0.275	−34056.5	−34037.6	0

file
mcmc_south_africa_step2_chain1_2026-10-19.Rdata
mcmc_south_africa_step2_chain2_2026-10-20.Rdata
mcmc_south_africa_step2_chain3_2026-10-20.Rdata
mcmc_south_africa_step2_chain4_2026-10-20.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):

age.20.to.29.transmission.multiplier.0	trate.3
1.936	1.920
age.30.to.39.transmission.multiplier.3	age.15.to.19.transmission.multiplier.0
1.891	1.807
trate.1	age.20.to.29.transmission.multiplier.2
1.797	1.718
age.30.to.39.transmission.multiplier.0	age.15.to.19.transmission.multiplier.3
1.696	1.691
age.20.to.29.transmission.multiplier.3	trate.4
1.639	1.531
trate.2	log.OR.testing.intercept
1.446	1.425
age.50.and.over.transmission.multiplier.3	female.to.male.multiplier
1.418	1.370
age.20.to.29.transmission.multiplier.1	trate.0
1.352	1.290
over.80.mortality.intercept.multiplier.male	age.30.to.39.transmission.multiplier.2
1.203	1.197
age.25.to.49.hiv.mortality.multiplier.2	age.25.to.49.hiv.mortality.multiplier.1
1.196	1.190
age.15.to.19.transmission.multiplier.1	age.0.to.14.hiv.mortality.multiplier.1
1.190	1.182
age.15.to.19.transmission.multiplier.2	hiv.specific.mortality.rates.1
1.176	1.173
age.40.to.49.transmission.multiplier.2	
1.163	

Merged simset:

location	simulations	total.weight	file
South Africa	1000	1000	simset_south_africa_step2_chains1234_2026-10-20.Rdata

South Africa step 2: likelihood

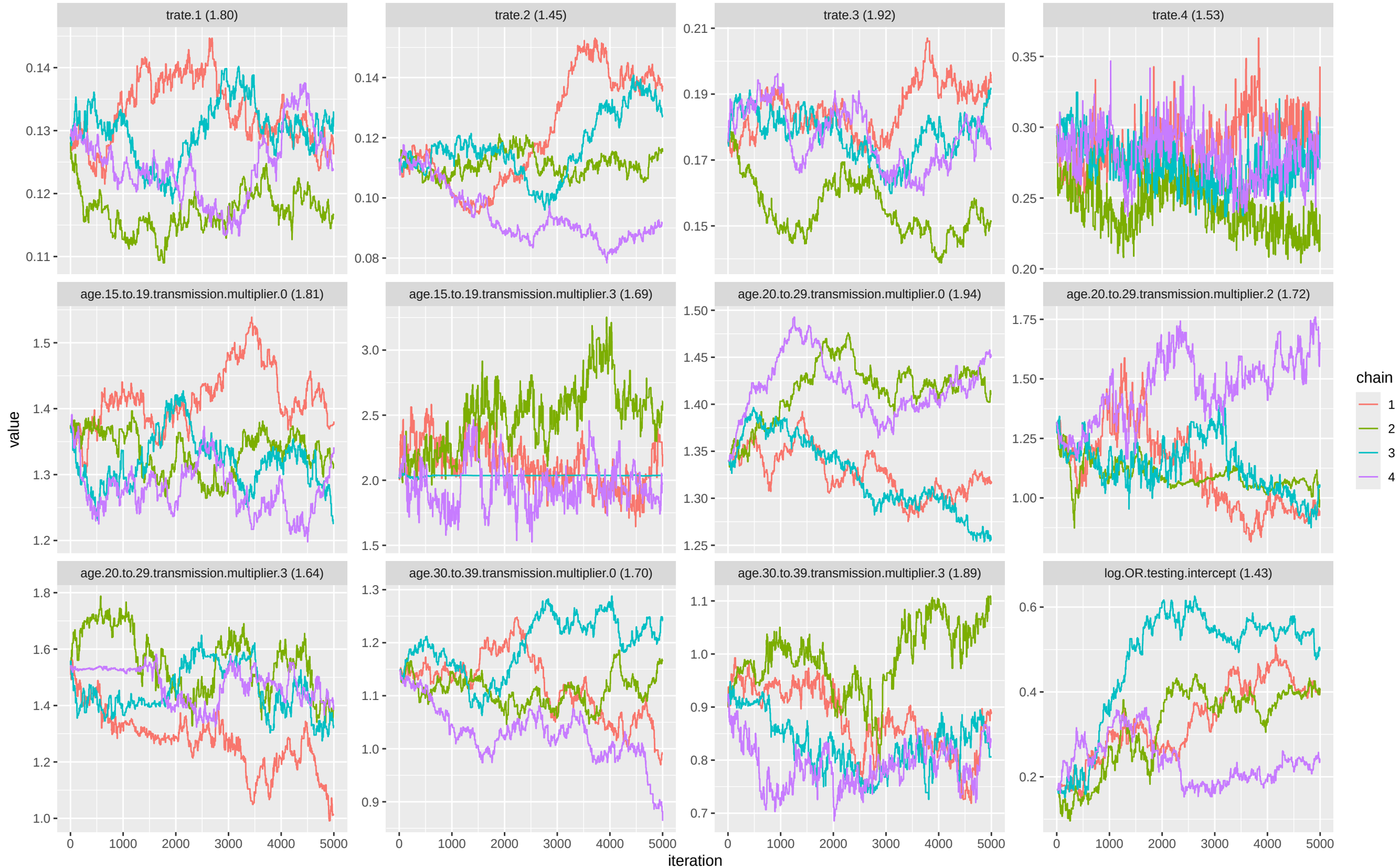
0
−10000
−20000
−30000
lue

0
25000
50000
75000
100000
iteration

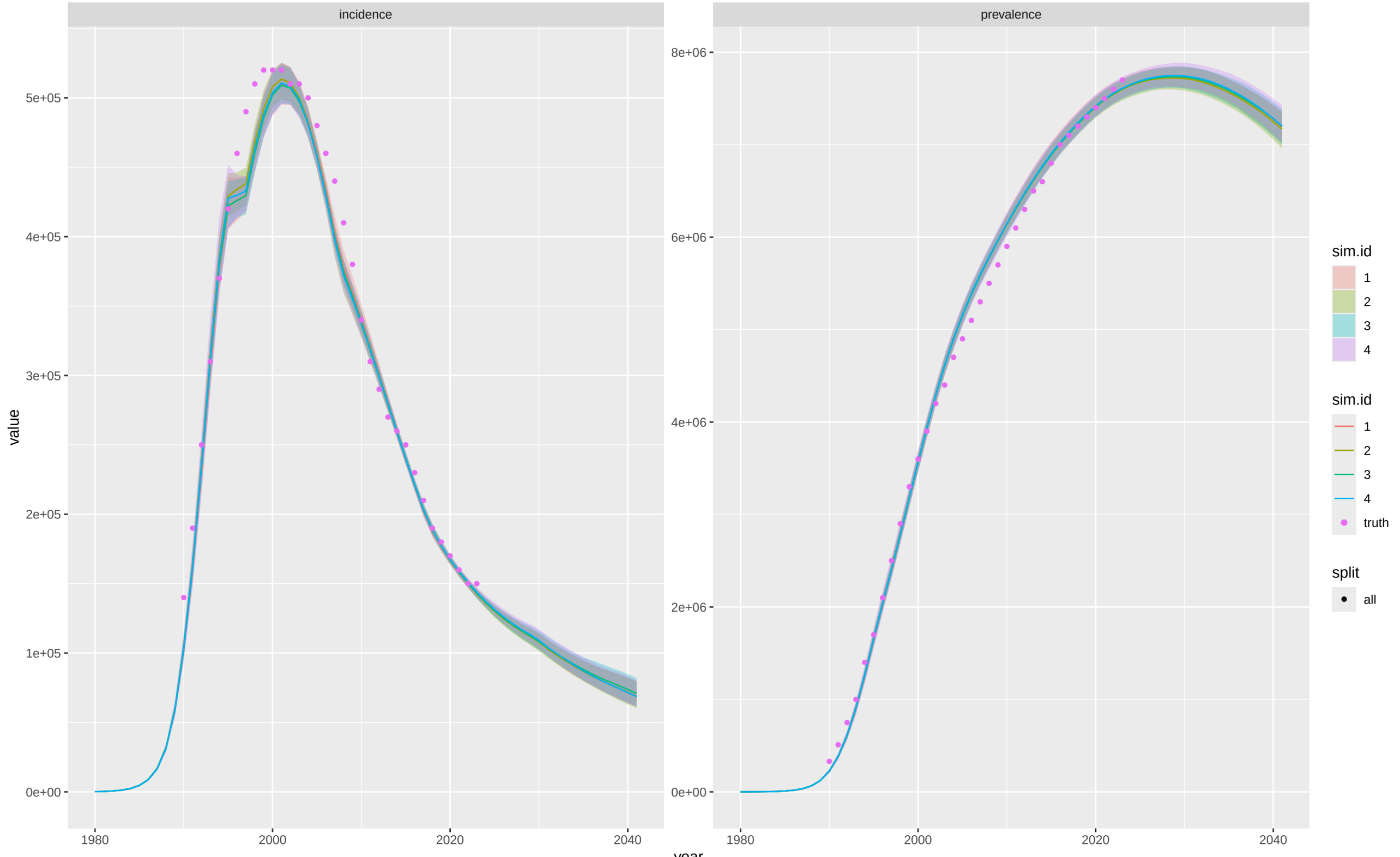
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

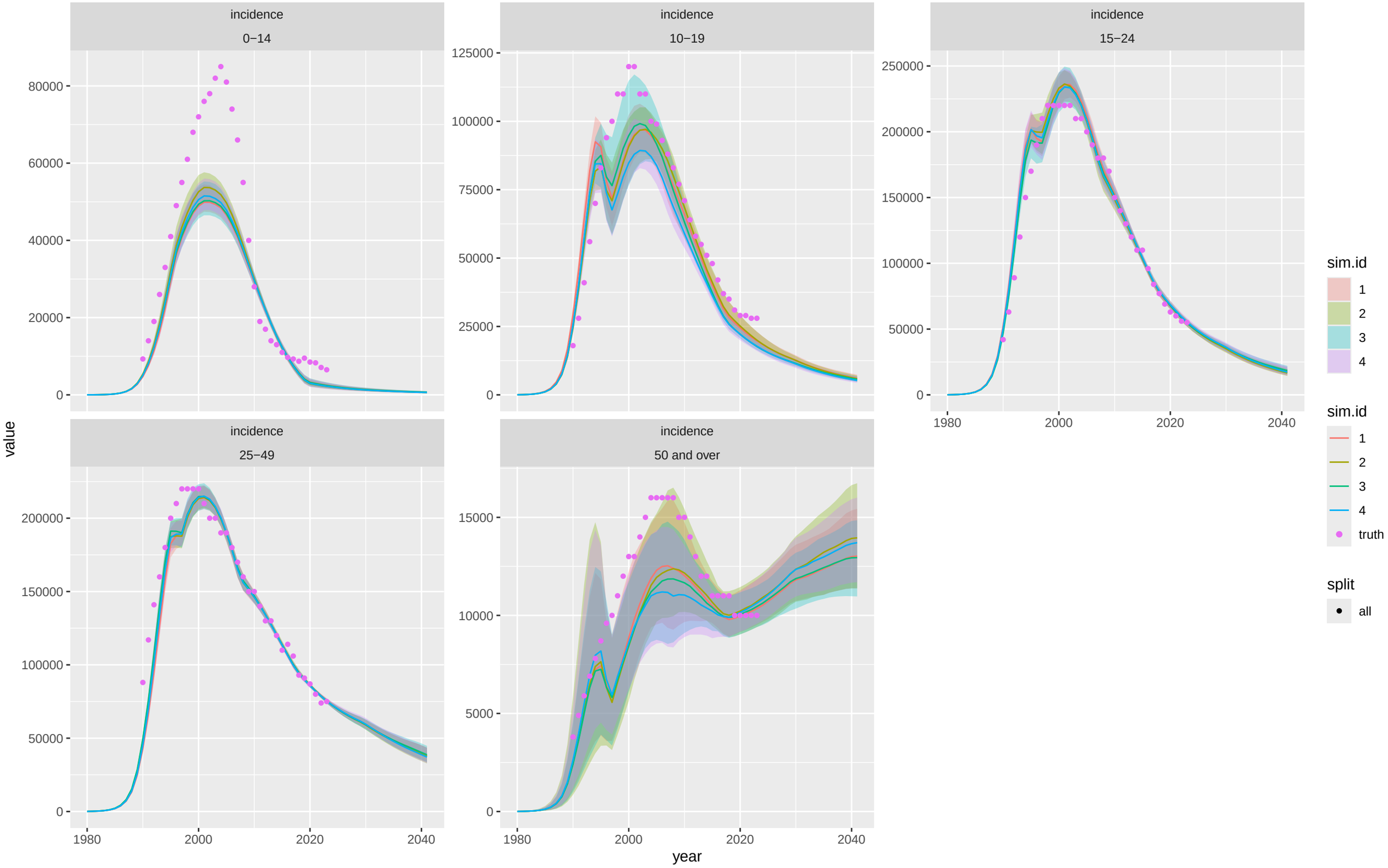
South Africa step 2: trace



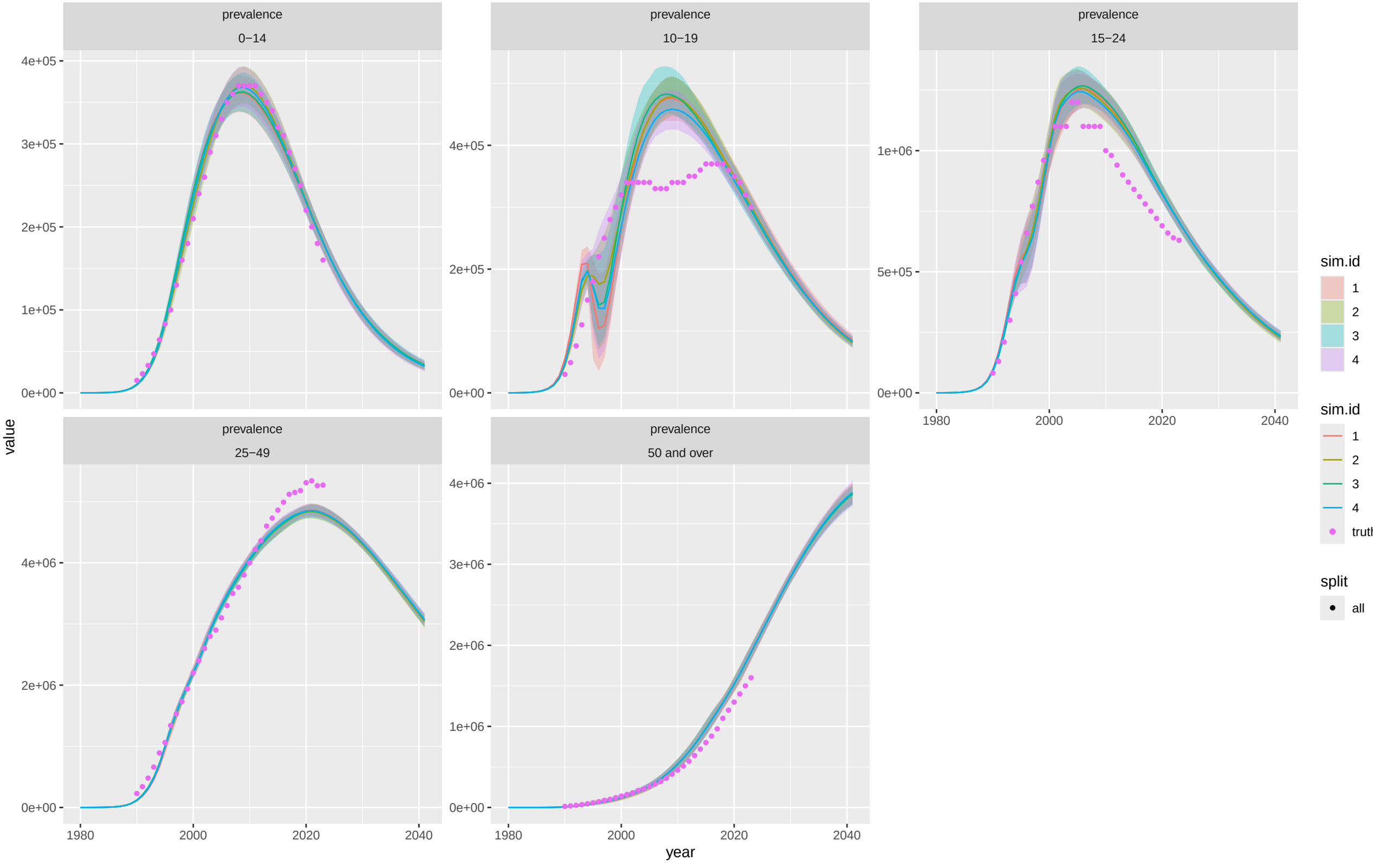
South Africa step 2: fit (one color per chain)



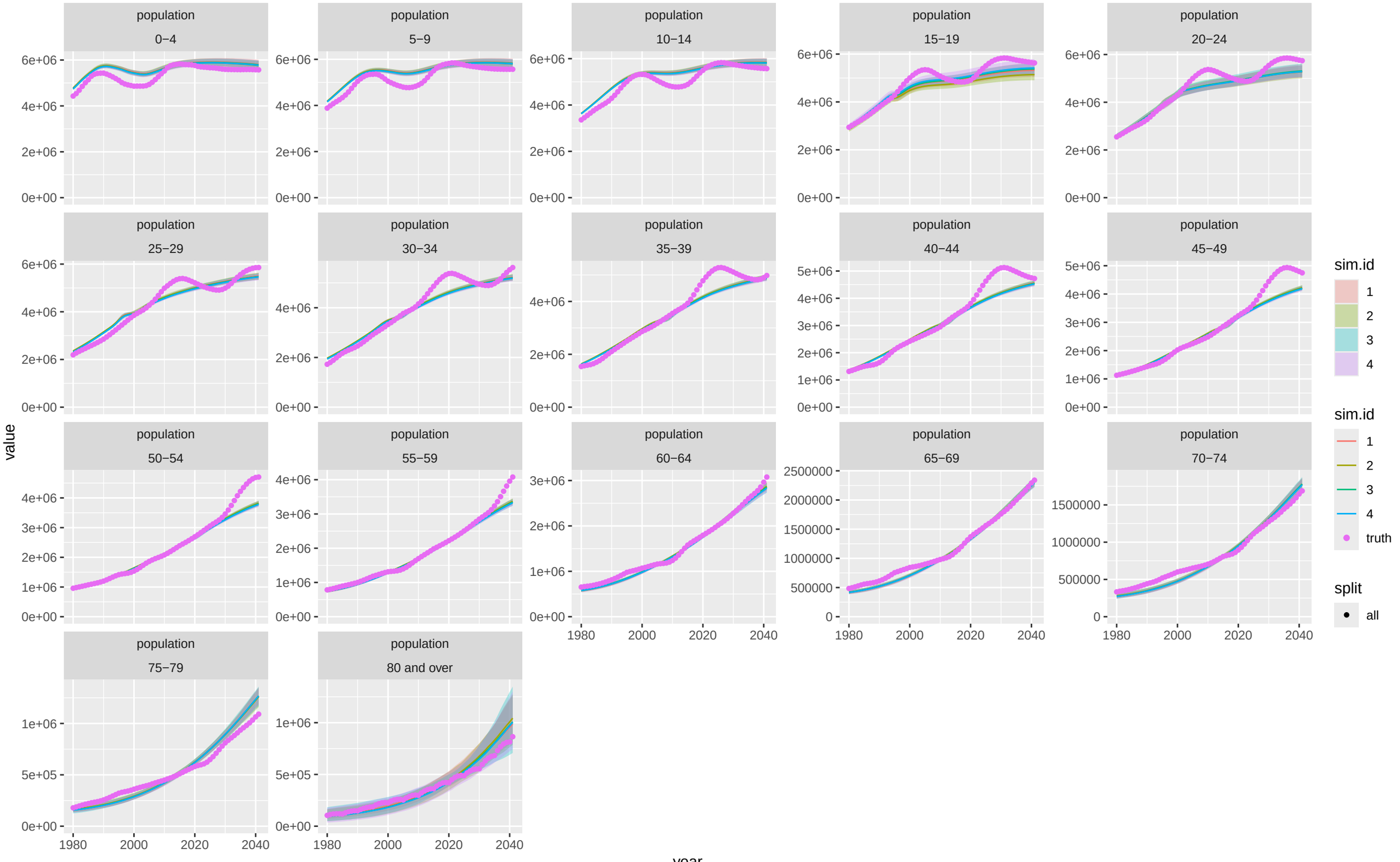
South Africa step 2: incidence.by.age (one color per chain)



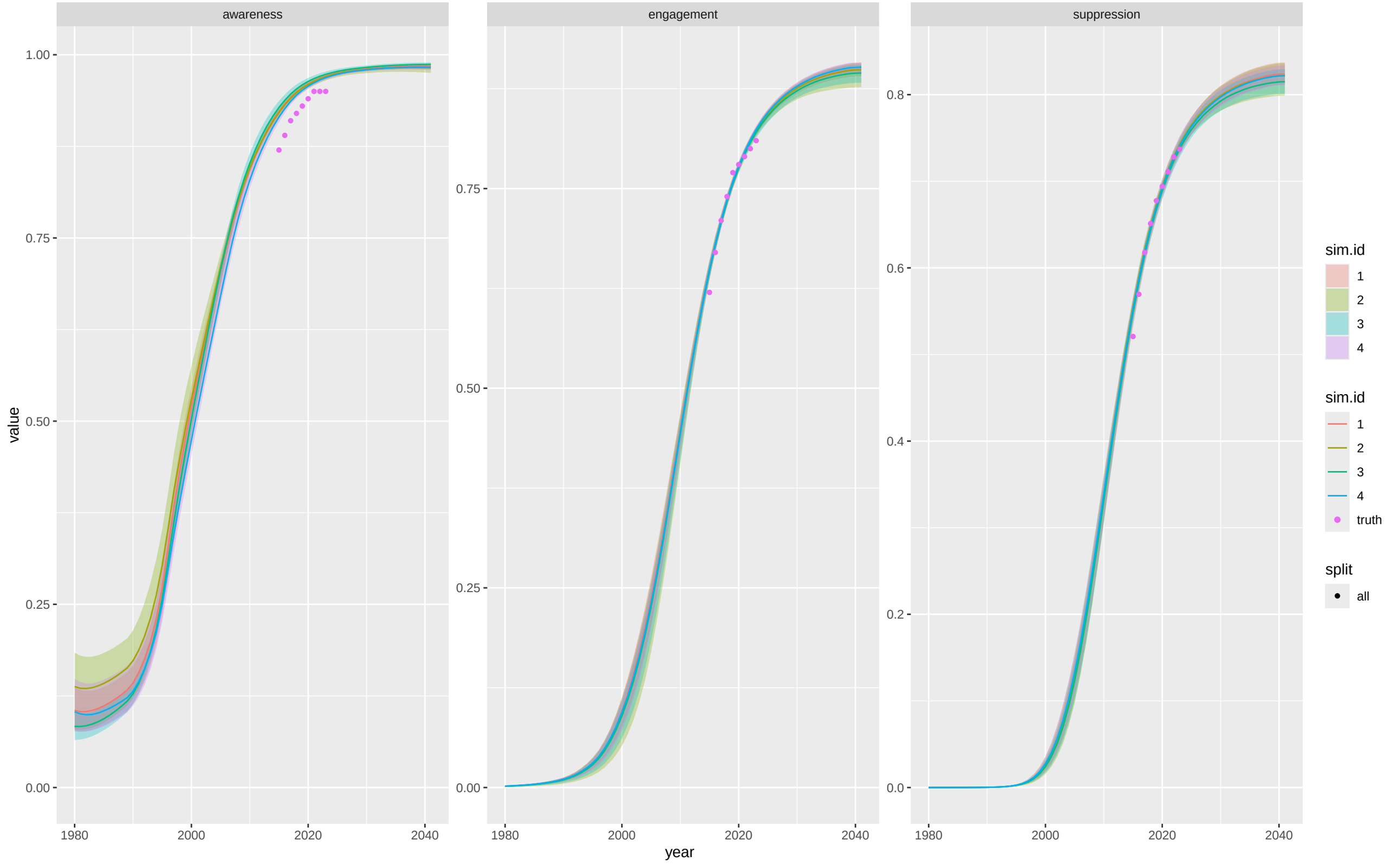
South Africa step 2: prevalence.by.age (one color per chain)



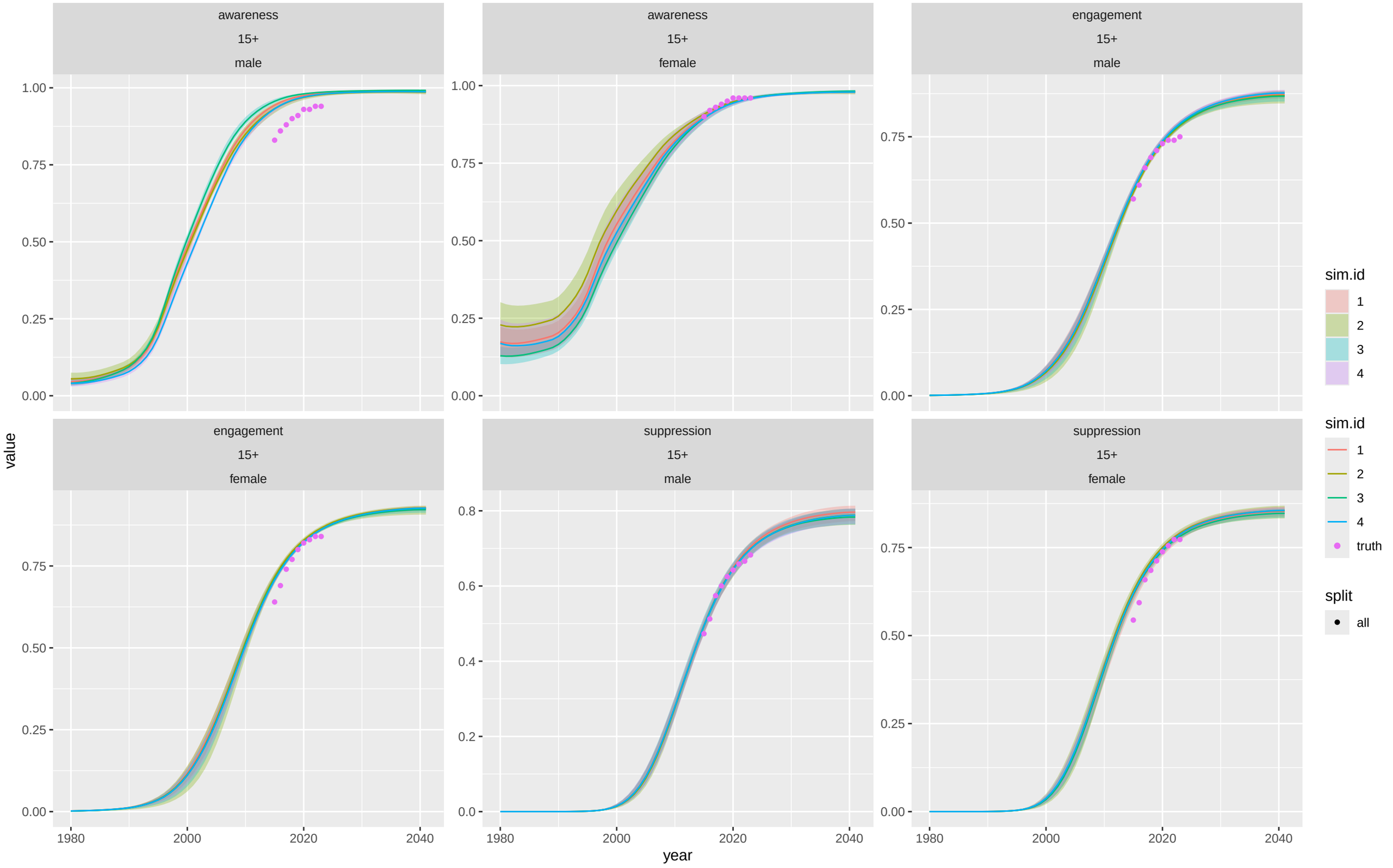
South Africa step 2: population.by.age (one color per chain)



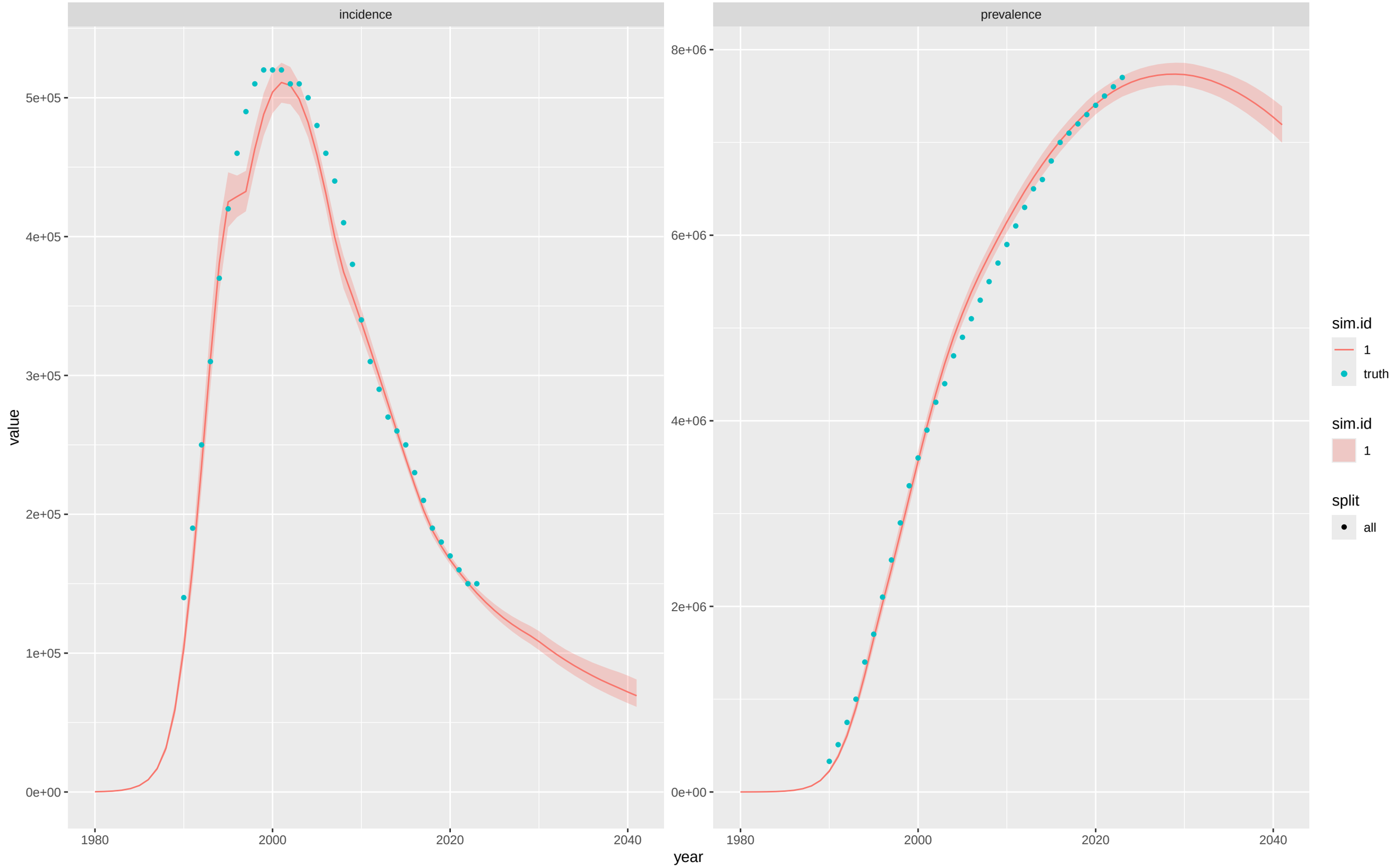
South Africa step 2: cascade (one color per chain)



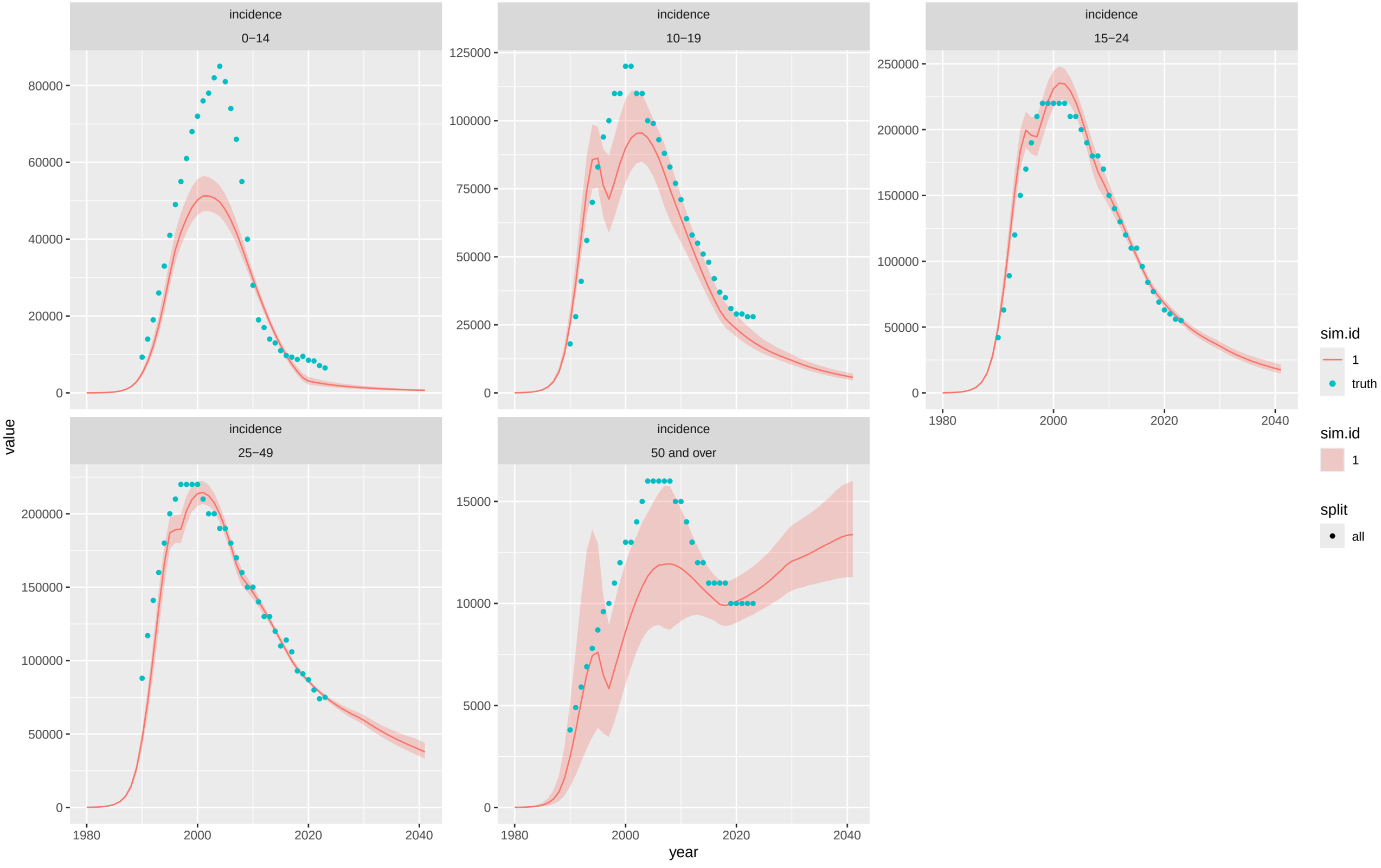
South Africa step 2: cascade.by.age.sex (one color per chain)



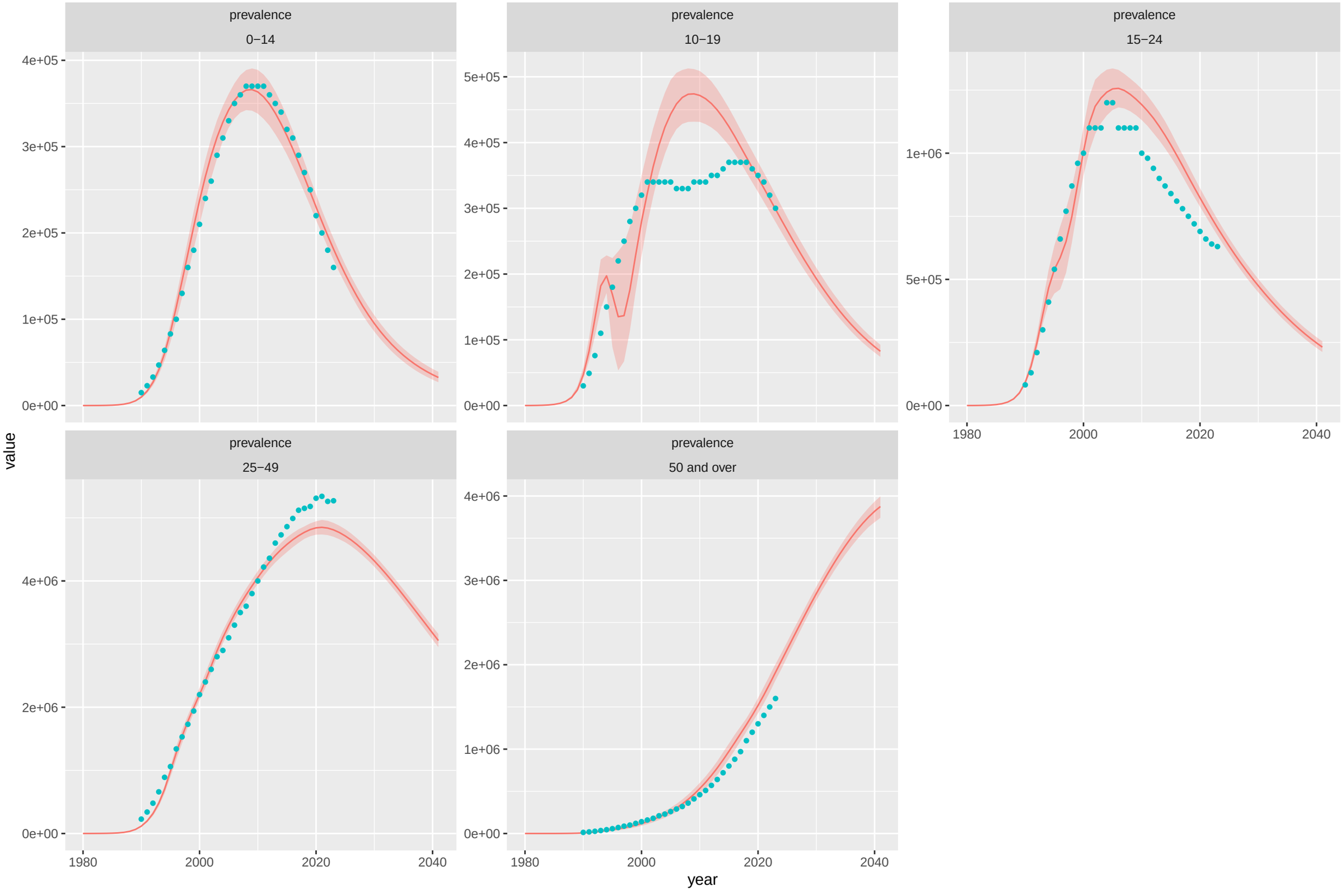
South Africa: fit (merged Step 2 simset)



South Africa: incidence.by.age (merged Step 2 simset)



South Africa: prevalence.by.age (merged Step 2 simset)



South Africa: population.by.age (merged Step 2 simset)

population 0−4
population 5−9
population 10−14
population 15−19
population 20−24
population 25−29
population 30−34
population 35−39
population 40−44
population 45−49
population 50−54
population 55−59
population 60−64
population 65−69
population 70−74
population 75−79
population 80 and over

value

year

sim.id
1
truth

split
all

South Africa: cascade (merged Step 2 simset)

awareness

engagement

suppression

value

year

1.00
0.75
0.50
0.25
0.00

0.8
0.6
0.4
0.2
0.0

1980
2000
2020
2040

sim.id
1
truth

sim.id
1

split
all

South Africa: cascade.by.age.sex (merged Step 2 simset)

